PROFESSIONAL INSTALLATION MANUAL | Ubuntu Server Deployment Using Oracle VirtualBox

ITEP 414 – System Administration and Maintenance

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Introduction

This manual provides a complete procedure for deploying Ubuntu Server 24.04 LTS in a VirtualBox virtual machine. The document is designed so that another system administrator can reproduce the installation without requiring additional instructions.

The manual includes virtual machine creation, operating system installation, initial server configuration, verification procedures, troubleshooting techniques, and recommended best practices for enterprise-style server deployment.

Ubuntu Server Installation Guide

Step 1 – Create the Virtual Machine

1. Open Oracle VirtualBox.
2. Click New.
3. Configure the following:

Setting	Value
Name	Ubuntu-Server-Week03
Type	Linux
Version	Ubuntu (64-bit)

Click Next.

Step 2 – Configure Memory and CPU

Memory

Set 4096 MB (4 GB).

Processor

1. Open Settings → System → Processor.
2. Set Processor(s): 2.

Step 3 – Create Virtual Hard Disk

1. Select Create a virtual hard disk now.
2. Choose VDI (VirtualBox Disk Image).
3. Select Dynamically allocated.
4. Set the size to 40 GB.
5. Click Create.

Step 4 – Attach Ubuntu Server ISO

1. Open Settings → Storage.

2. Select Empty under Controller: IDE.
3. Click the CD icon.
4. Choose Choose a disk file.
5. Select the Ubuntu Server ISO.
6. Click OK.

Step 5 – Start the Installer

1. Click Start.
2. Select Try or Install Ubuntu Server.
3. Choose Install Ubuntu Server.

Step 6 – Configure Installation Options

Use the following values:

Option	Value
Language	English
Keyboard	English (US)
Network	DHCP
Hostname	(server name)
Username	(preferred username)
Disk Setup	Guided – Use Entire Disk
File System	ext4
Install OpenSSH Server	Enabled

Step 7 – Complete Installation

1. Wait for the installation to finish.
2. Select Restart Now.
3. Remove the installation ISO when prompted.
4. Press Enter to continue booting.

Expected Result: The system displays the Ubuntu login prompt.

Initial Server Configuration

After rebooting, log in using the created account.

Verify Hostname

hostname

Expected output:

(server name)

Verify IP Address

ip addr

Look for a line similar to:
inet 10.0.2.15/24

Test Internet Connectivity
ping -c 4 8.8.8.8
ping -c 4 google.com

Update the Server
sudo apt update
sudo apt upgrade -y

Install OpenSSH Server (if missing)
sudo apt install openssh-server -y
sudo systemctl enable ssh
sudo systemctl start ssh

Server Verification Procedures

Verification Task	Command	Expected Result
Verify hostname	hostname	(server name)
Verify IP address	ip addr	Displays a valid IP address
Test internet	ping -c 4 google.com	Successful replies received
Update repositories	sudo apt update	No repository errors
Verify SSH service	systemctl status ssh	active (running)

Troubleshooting Guide

Problem 1 – No Internet Connection
Symptoms

- ping google.com fails
- apt update cannot connect to repositories

Solution

1. Shut down the VM.
2. Open VirtualBox → Settings → Network.
3. Ensure:

Setting	Value
Enable Network Adapter	Checked
Attached to	NAT

Restart the virtual machine

Problem 2 – Failed to fetch Error During apt update
Cause
The default Ubuntu mirror may be unavailable or blocked.

Solution
Edit the repository source:
sudo nano /etc/apt/sources.list.d/ubuntu.sources
Replace:
http://ph.archive.ubuntu.com/ubuntu
with:
http://archive.ubuntu.com/ubuntu

Save the file and run:
sudo apt clean
sudo apt update

Problem 3 – SSH Service Not Found
Symptoms
ssh.service could not be found

Solution
Install OpenSSH Server:
sudo apt install openssh-server -y
sudo systemctl enable ssh
sudo systemctl start ssh

Verify the service:
systemctl status ssh

Best Practices

- **Use Non-Root Administrative Accounts**
  Create a standard user with sudo privileges instead of logging in directly as root.

- **Keep the System Updated**
  Regularly apply security and package updates:

  sudo apt update
  sudo apt upgrade -y

- **Secure SSH Access**
  1. Use strong passwords.
  2. Change default credentials immediately.
  3. Consider using SSH key authentication for remote administration.

- **Create VirtualBox Snapshots**
  Before performing major configuration changes:
  1. Open VirtualBox Manager.
  2. Select the VM.
  3. Click Snapshots → Take.
  4. Provide a descriptive snapshot name.

- **Maintain Proper Documentation**
  Record:
  1. Hostname
  2. IP address
  3. Installed packages
  4. Configuration changes
  5. Troubleshooting actions performed

  Proper documentation improves system maintenance, troubleshooting efficiency, and administrative consistency.

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